

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Pseudocode Writing Task (Algorithm Design)

Weightage: 20%

QUESTIONS:

Total Marks: 25

1: XML Parsing Algorithm

Write a detailed **pseudocode algorithm** to parse an XML document that contains structured data such as user or product details. Your algorithm should clearly describe the steps involved in loading the XML file, accessing its elements, traversing through nodes, extracting required data, and displaying the output in a readable format. Ensure that the logic handles hierarchical structure properly.

Answer:

START

INPUT XML_File

LOAD XML_File into XML_Document

IF XML_Document is not loaded THEN

 DISPLAY "Error: Unable to load XML file"

 STOP

END IF

GET Root_Node from XML_Document

DISPLAY "Root Element: ", Root_Node.Name

FOR EACH Child_Node in Root_Node

 DISPLAY "Record Details"

 FOR EACH Element in Child_Node

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    GET Element_Name
    GET Element_Value

    IF Element has Child_Nodes THEN
        FOR EACH Sub_Element in Element
            DISPLAY Sub_Element.Name, ": ", Sub_Element.Value
        END FOR
    ELSE
        DISPLAY Element_Name, ": ", Element_Value
    END IF

END FOR

DISPLAY "-----"

END FOR

CLOSE XML_Document

STOP
Flow:
Load XML file → Access root element → Traverse parent and child nodes → Extract element values → Handle nested elements → Display data.

```

2: MVC Workflow

Write a **pseudocode representation** of the MVC (Model–View–Controller) workflow in a web application. Your answer should clearly illustrate how a user request is received, processed by the Controller, how the Model interacts with the database, and how the View generates and returns the response to the user. Include all major steps involved in request handling and response generation.

Answer:

```

START

USER sends Request through Browser

WEB SERVER receives Request

ROUTE Request to CONTROLLER

CONTROLLER processes Request

VALIDATE Request Data

IF Request Data is invalid THEN
    SEND Error Response to VIEW
ELSE

```

CONTROLLER calls MODEL

MODEL processes Business Logic

MODEL connects to DATABASE

EXECUTE Database Operation

RECEIVE Data from DATABASE

MODEL returns Data to CONTROLLER

CONTROLLER sends Data to VIEW

VIEW generates Dynamic Web Page

SERVER sends Generated Response to USER

END IF

STOP

Flow:

User → Controller → Model → Database → Model → Controller → View → User.

3: JSP Request Flow

Write a detailed **pseudocode algorithm** to explain the request–response flow in a JSP-based web application. Your algorithm should include steps such as client request initiation, server processing, interaction with backend components (like Servlets or database), and generation of dynamic content that is sent back to the client.

Answer:

START

CLIENT opens Web Browser

CLIENT sends Request for JSP Page

WEB SERVER receives JSP Request

CHECK whether JSP is already compiled

IF JSP is not compiled THEN

 TRANSLATE JSP into Servlet

 COMPILE Servlet

END IF

EXECUTE JSP Servlet

JSP receives Request Data

IF Backend Processing is required THEN

CALL Servlet or Java Backend Component

Backend Component processes Request

CONNECT to DATABASE if required

EXECUTE Database Query

RECEIVE Result

RETURN Result to JSP

END IF

JSP processes Received Data

GENERATE Dynamic HTML Content

SERVER creates HTTP Response

SEND Response to CLIENT

CLIENT Browser displays Web Page

STOP

Flow:

Client → JSP Request → Server → JSP/Servlet Processing → Backend/Database → JSP → Dynamic HTML → Client.

4: Database Retrieval

Write a **pseudocode algorithm** to retrieve data from a database. Your answer should include steps for establishing a database connection, executing SQL queries, processing the result set, and displaying the retrieved data. Clearly represent the sequence of operations involved in database interaction.

Answer:

START

LOAD Database Driver

CREATE Database Connection

IF Connection fails THEN

DISPLAY "Database Connection Failed"

STOP

```

END IF

CREATE SQL Query

EXECUTE SQL Query

STORE Result in Result_Set

IF Result_Set is empty THEN
    DISPLAY "No Records Found"
ELSE

    WHILE Next Record exists in Result_Set

        READ Required Columns

        DISPLAY Retrieved Data

    END WHILE

END IF

CLOSE Result_Set

CLOSE Database Connection

DISPLAY "Data Retrieval Completed Successfully"

STOP

```

Flow:

Load Driver → Connect Database → Execute SQL Query → Get Result Set → Process Records → Display Data → Close Connection.

5: PHP Validation

Write a **pseudocode algorithm** for validating user input in a web form using PHP concepts. The algorithm should include checks for empty fields, validation of input formats (such as email and password), handling invalid inputs with appropriate error messages, and allowing submission only when all validations are satisfied.

Answer:

```

START

USER enters Data in Web Form

USER clicks Submit Button

```

CHECK whether Form is submitted

IF Form is submitted THEN

 GET Name, Email, Password and other Input Fields

 SET Validation_Status = TRUE

 IF Name is empty THEN

 DISPLAY "Name cannot be empty"

 SET Validation_Status = FALSE

 END IF

 IF Email is empty THEN

 DISPLAY "Email cannot be empty"

 SET Validation_Status = FALSE

 ELSE IF Email format is invalid THEN

 DISPLAY "Enter a valid email address"

 SET Validation_Status = FALSE

 END IF

 IF Password is empty THEN

 DISPLAY "Password cannot be empty"

 SET Validation_Status = FALSE

 ELSE IF Password length is less than required length THEN

 DISPLAY "Password is too short"

 SET Validation_Status = FALSE

 END IF

 IF Validation_Status = TRUE THEN

 SANITIZE User Input

 PROCESS Form Data

 DISPLAY "Form Submitted Successfully"

 ELSE

 DISPLAY "Please correct the errors and try again"

 END IF

END IF

STOP

Flow:

User Input → Check Empty Fields → Validate Email → Validate Password → Show Errors if Invalid → Process and Submit if Valid.

Code of Conduct:

I P.Harika(192311243) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Harika

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient